

Meaningful Use to Secure Interoperability

Addressing Cybersecurity Risks in Healthcare

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The Health Information Technology for Economic and Clinical Health (HITECH) Act promoted the adoption of electronic health records (EHRs), while the Meaningful Use program provided financial incentives beginning in 2011 to further expand EHR adoption and technology use. Meaningful Use encouraged the use of technologies such as clinical decision support (CDS), computerized provider order entry (CPOE), electronic prescribing (e-prescribing), and health information exchange (HIE), improving access to health information and contributing to healthcare quality (Centers for Medicare & Medicaid Services [CMS], 2023; Kan, 2011; Jones et al., 2014).

However, the benefits of greater connectivity also created broader cybersecurity responsibilities. As healthcare organizations connected more systems and exchanged information across clinical environments, the number of potential access points and dependencies requiring protection increased (Office of the National Coordinator for Health Information Technology [ONC], 2026; Coventry & Branley, 2018). Although expanding access to information can improve clinical efficiency and coordination, it also requires organizations to secure increasingly interconnected environments. HITECH and the Meaningful Use program accelerated healthcare technology adoption while expanding the cybersecurity responsibilities associated with interoperability.

Fast Healthcare Interoperability Resources (FHIR) illustrates how increased interoperability can create new cybersecurity challenges by enabling API-based communication among EHR platforms, mobile applications, patient portals, and other healthcare technologies (Health Level Seven International [HL7], 2023; Saripalle et al., 2019). Greater connectivity can increase the potential impact of authorization failures involving sensitive health information. For example, CVE-2026-24487 affected the OpenEMR FHIR CareTeam endpoint and allowed

patient-scoped tokens to access care team data belonging to other patients (National Institute for Standards and Technology [NIST], 2026; SentinelOne, 2026). This vulnerability demonstrates why authentication alone is insufficient. An authenticated user or application may still be improperly authorized to access information belonging to another patient. Organizations must verify that each API request is authorized to access the specific patient information permitted by the user's scope. Encryption mechanisms such as TLS can protect FHIR data during transmission, but they do not determine whether an authenticated user is authorized to access a particular patient's information.

Digital Imaging and Communications in Medicine (DICOM) provides another example of how interoperability can introduce cybersecurity risks within healthcare environments (National Electrical Manufacturers Association [NEMA], n.d.). DICOM-enabled imaging systems have improved teleradiology by allowing radiologists to interpret images remotely and access results more efficiently (CARE, 2025). However, Black Hat research identified 18.2 million patient records in the United States associated with internet-accessible DICOM servers, exposing these systems to ransomware and other attacks that can disrupt healthcare delivery and expose patient information (Desjardins et al., 2020; Yazdanmehr, 2023a, 2023b). Internet-exposed DICOM servers can allow attackers to modify images or introduce malicious code that can spread through PACS and connected systems. Legacy systems create additional risk because they may continue supporting essential clinical functions while introducing vulnerabilities into newer technologies connected to them. This illustrates how a vulnerability in one system can create security risks across an interconnected healthcare environment. The risk extends beyond unauthorized access because altered diagnostic images can compromise information integrity. If clinicians rely on corrupted images, they may make incorrect diagnoses or treatment decisions,

potentially resulting in serious adverse patient outcomes. Similarly, when ransomware makes patient records, diagnostic results, or communication systems unavailable, clinicians may have to delay treatment or make decisions with incomplete information. This demonstrates how cybersecurity incidents can directly disrupt clinical operations, potentially resulting in patient harm or even death (Computer Fraud & Security, 2020).

The FHIR and DICOM examples demonstrate why interoperability requires layered security controls addressing unauthorized access, data manipulation, and system disruption. Because a compromise in one environment can affect interconnected clinical systems, organizations should prioritize these controls based on their potential impact on clinical operations and patient safety.

Since compromised data and unavailable systems can result in patient harm, cybersecurity should be treated as part of patient safety and clinical risk management rather than solely as a technical or compliance function. Preventive controls such as zero trust principles, session timeouts, and multi-factor authentication (MFA) can strengthen authentication and help ensure that access requests are appropriately verified. Role-based access control (RBAC) further strengthens security by limiting users' access to the information necessary for their intended functions. Encryption of data in transit and at rest helps protect sensitive information from unauthorized disclosure, while additional integrity controls can help detect unauthorized modification of data. API governance and network segmentation can further limit unauthorized access and contain the effects of a compromise across interconnected clinical systems. Effective downtime procedures can also reduce the impact of disruptions to systems that healthcare professionals depend on for clinical decision-making (Vukotich, 2023).

These architectural concerns also extend to AI-enabled clinical technologies, which increasingly rely on APIs, cloud services, and third-party applications to process healthcare information. The NIST CSF 2.0 and AI RMF 1.0 provide complementary approaches for managing cybersecurity and AI-specific risks. However, the underlying challenge remains consistent with FHIR and DICOM: healthcare organizations must secure information as it moves across technological and organizational boundaries rather than protecting individual systems in isolation.

In conclusion, Meaningful Use improved access to health information and encouraged greater interoperability, but the resulting expansion of connectivity also created more complex security responsibilities. The FHIR and DICOM examples demonstrate why cybersecurity must evolve alongside interoperability while addressing both legacy technical debt and emerging technologies. Although the HIPAA Security Rule requires administrative, physical, and technical safeguards to protect ePHI, it does not prescribe specific technologies or a particular cybersecurity framework (U.S. Department of Health & Human Services [HHS], n.d.). Healthcare organizations must therefore apply risk-based practices such as system patching, legacy modernization, risk management, and vendor oversight to achieve secure interoperability beyond basic HIPAA compliance. Future interoperability initiatives should evaluate security maturity alongside connectivity rather than treating cybersecurity as a secondary consideration. Ultimately, interoperability should be measured not only by how much information can be exchanged, but also by whether that information and the systems that deliver it can be trusted when patient care depends on them.

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